

Diwali Sales Data Analysis – Customer Insights for Festive Retail

Objective:

To explore consumer behaviour during the Diwali shopping season using Python. This end-to-end data analysis project reveals purchasing patterns, high-performing customer segments, and product trends to help retailers strategize more effectively during festivals.

Tools & Technologies

- **Languages & Libraries:** Python, Pandas, NumPy
 - **Visualization:** Matplotlib, Seaborn
 - **Platform:** Jupyter Notebook
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Project Workflow

◆ Data Acquisition & Cleaning

- Imported a raw CSV dataset of Diwali transactions.
- Removed null values and unnecessary fields (e.g., Status, unnamed1).
- Converted key fields like Amount for accurate numeric analysis.

◆ Exploratory Data Analysis

- Conducted univariate and multivariate analysis on demographics (Gender, Age, Marital Status, Occupation).
 - Explored city-tier distribution and product category performance.
 - Created intuitive visualizations to uncover trends and compare customer segments.
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Key Business Insights

Who Shops the Most?

- **Women aged 26–35** emerged as the most active buyers during Diwali.

Where is the Market?

- **Tier 1 cities** like Mumbai, Delhi, and Bangalore generated the highest revenue, indicating urban buying power.

What Do They Do?

- **Working professionals**, especially in IT, were top contributors to overall sales.

What Are They Buying?

- **Apparel** and **electronics** dominated sales volumes, driven by festive discounts.

Strategic Recommendations:

- Focus on metro-centric marketing.
- Target female professionals in the 26–35 age group.
- Push promotions for top-performing categories like gadgets and fashion.

Conclusion

The Diwali sales analysis highlights that young female professionals in Tier 1 cities are the most active shoppers, with apparel and electronics as the top-selling categories. Targeted marketing toward this demographic, coupled with a focus on metro-based campaigns and high-value customer engagement, can significantly boost festive sales. These insights provide a clear direction for optimizing future Diwali and seasonal strategies through data-driven decision-making.